Emile Gaudinot

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Education

- **Technical University of Berlin** — Berlin, Germany
  Applied Mathematics, Statistics, Data Science (MSc), GPA: 4.0 — *2022 – 2023*
- **École Centrale de Nantes** — Nantes, France
  Engineering Degree (MSc), GPA: 4.0 — *2020 – 2022*

Experience

- **Sia** — Paris, France
  Experienced Data Science Consultant — *Feb. 2026 – Present (9 months)*
  - **Electrical Network RL Optimization (Major Energy Provider)**: Using **Reinforcement Learning (PPO)** and a **GNN**, optimized the critical operations on the Electricity Network. **+500%** speed enabled real-time use.
  - **Revenue Forecast (Luxury Client)**: Trained a **Holt-Winters** model on **GCP Vertex AI**. Predicted daily sales over 12 months for each of the 600 stores in the world. **-17%** WMAPE allowed deployment in production.
  - **Video Moderation System (Media Client)**: Automated the blurring of alcohol, cigarettes and fireguns with **YOLO26**/**RF-DETR** models on **AWS SageMaker**. **React** dashboard generated from a **Langflow** agentic pipeline. App fully deployed on **Databricks**.
  - **Pengwin Challenge (World-Wide Medical Challenge)**: Automated the segmentation of bone fragments on pelvis fracture **CT-scans** with 2 **nnUNet**. Reduction planning with **PointNet** + **Transformer**. Top 100 world.
- **Fraunhofer - Heinrich-Herz-Institute** — Berlin, Germany
  Experienced Data Scientist — *Mar. 2025 – Feb. 2026 (1 year)*
  - **Perceived Distance Prediction from Brain Signals**: Developed an innovative AI pipeline using **MNE-Python** with (1) an **SSD** and a **SPoC regression** for distance, and (2) a **ROC classifier** for personal space intrusion detection (F1 = 0.81). Extension of the pipeline for **real-time** using **LSL** (F1 = 0.65) and deployed with **Databricks Workflows**. This project finally brought the research field to life. Published the results in a research paper.
- **Charité** — Berlin, Germany
  Data Scientist — *Mar. 2023 – Feb. 2025 (2 years)*
  - **Computer Vision for Early Detection of Coronary Artery Disease**: Segmented arteries from 3D heart images with **U-Net**, **U-net Transformer** and **MedSAM**, on **MONAI** (IoU = 0.79). Implemented the pipeline for artery graph extraction with **scikit-image** (**CCA**, **skeletonization**, etc). Monitoring with **MLflow**. Released in production with **Databricks**, leading to **+60%** correct CAD diagnosis. Published the results in a research paper.
  - **Intelligent Computer Security**: (1) Detected malwares for .pdf and .docx using **Bag-of-Words**, **PCA** and **XGBoost** (F1 = 0.98) (2) Detected network intrusions using **Isolation Forests** and **LOF scores** (**-80%** false positives) (3) Robustified models to **Adversarial Attacks** using **i-FGSM**. Deployed to protect Charité's network.

Projects

- **Winner of BIFOLD's Cybersecurity Hackathon (Oct. 2023 – Mar. 2024)**: Trained and fine-tuned ML models on 4 cybersecurity challenges. Price Ceremony (1st Price/52) at the German Ministry of Research, Technology and Space.
- **Local LLM - Complete app (Apr. 2025 – Nov. 2025)**: Developed and optimized a custom ChatBot application which runs locally. Possibility to upload documents (img, pdf, docx, pptx) thanks to an OCR + text extraction pipeline.

Skills

- **LLMs**: Mistral API, Ollama, Light local models (ministral-3:8b, gemma4-e4b, llama3.1-8b)
- **Deep Learning**: RL (ActorCritic, PPO), GNN (GAT, GCN, GraphSAGE), Time series (RNN, LSTM), Vision (nnU-Net, UNETR, MedSAM, YOLO26, RF-DETR)
- **Tools**: Production (AWS, GCP, Databricks, Docker, MLflow, Git), Python (PyTorch, Scikit-learn, Pandas), HuggingFace
- **Programming Languages**: Python, SQL, Bash, C++, R
- **Spoken Languages**: French (native), English (C1 - TOEFL 104/120), German (C1 - ZEMS 92/100)

Certifications: Databricks Certified Machine Learning Associate, ANSSI SecNum: Cybersecurity & Digital Tool Protection